

HP PAVILION

PRICE COMPARISON TOOL

Project Documentation & Technical Report

- Web Scraping
- Data Analysis
- Visualization
- Statistics
- Business Insights

Project Title	HP Pavilion Laptop Price Comparison Tool
Project Type	Data Analytics & Web Scraping Project
Technology Stack	Python, Selenium, Pandas, NumPy, SciPy, Matplotlib, Seaborn
Data Sources	Amazon India & Flipkart (Live web scraping)
Target Product	HP Pavilion Laptop Series
Document Version	v1.0 — Final
Purpose	Client Submission LinkedIn Portfolio GitHub Repository

1. Executive Summary

The HP Pavilion Price Comparison Tool is a data analytics solution designed to collect, clean, analyze, and visualize HP Pavilion laptop pricing data across two of India's largest e-commerce platforms — Amazon India and Flipkart. The project demonstrates an end-to-end data pipeline: from automated live data collection through web scraping to business-ready visual insights.

The tool enables data-driven purchasing decisions by answering key business questions:

- Which platform offers better prices for HP Pavilion laptops?
- Is the price difference between platforms statistically significant?
- How do customer ratings correlate with pricing?
- Which products offer the best value-for-money?

Key Outcome

By combining Selenium-based web scraping with statistical analysis and visualization, this project delivers actionable market intelligence that supports pricing strategy, competitor monitoring, and customer-focused product decisions.

The pipeline is fully reproducible and extensible to additional products or platforms.

2. Project Objectives

This project was built to achieve the following objectives:

- Automate real-time data collection from Amazon India and Flipkart using Selenium WebDriver.
- Clean and preprocess raw scraped data — handling missing values, currency symbols, and format inconsistencies.
- Perform exploratory data analysis (EDA) to understand price distributions, rating patterns, and outliers.
- Conduct statistical hypothesis testing (t-test) to verify whether platform price differences are statistically meaningful.
- Calculate confidence intervals to quantify pricing ranges for each platform.
- Visualize key metrics through six comprehensive charts for clear stakeholder communication.
- Identify and rank the most expensive, best-rated, and best value-for-money products.
- Generate business insights that support pricing strategy and competitive positioning.

3. Technology Stack

Category	Libraries / Tools
Web Scraping	Selenium WebDriver, ChromeDriverManager, WebDriverWait
Data Manipulation	Pandas, NumPy
Statistical Analysis	SciPy (ttest_ind, t.interval, sem), Pearson Correlation
Data Visualization	Matplotlib, Seaborn
Development Language	Python 3.x
Development Environment	Jupyter Notebook (.ipynb)
Data Export	CSV (hp_pavilion_price_comparison.csv)
Output	hp_pavilion_analysis.png — 2×3 subplot visualization

4. Project Workflow & Pipeline

The project follows a structured six-phase data pipeline:

#	Phase	Description
1	Browser Setup	Chrome WebDriver initialized with custom user-agent to bypass bot detection. Options include maximized window and disabled automation control flags.
2	Data Scraping	Automated scraping of up to 50 HP Pavilion product listings each from Amazon India and Flipkart. Extracts Product Name, Price, and Rating per listing using XPath selectors.
3	Data Merging & Filtering	Both datasets are combined via <code>pd.concat</code> . Rows are filtered to retain only records with 'HP' in the product name, removing unrelated results.
4	Data Cleaning	Price columns are cleaned: ₹ symbol and commas are stripped, values converted to float. Rows with unresolvable prices are dropped.
5	EDA & Statistics	Descriptive stats, confidence intervals, standard deviation comparison, t-test, Pearson correlation, price segmentation, and top-product rankings are computed.
6	Visualization & Export	Six-panel matplotlib/seaborn figure generated and saved. Data exported to CSV. Business insights and recommendations documented inline.

5. Web Scraping Design

5.1 Browser Configuration

Chrome is launched with the following anti-bot-detection settings:

- Custom User-Agent string mimicking a real Chrome 120 browser on Windows 10.
- AutomationControlled feature disabled to prevent detection by website security.
- Maximized window mode for consistent DOM rendering.
- ChromeDriverManager used for automatic driver version management.

5.2 Amazon Scraping

Target URL: <https://www.amazon.in/s?k=hp+pavilion>

- Searches for HP Pavilion laptops on Amazon India.
- Locates product containers using data-component-type='s-search-result' XPath.
- Extracts product name (a-size-medium a-color-base class), price (a-price-whole class), and rating (a-size-small a-color-base class).
- Collects up to 50 results per scraping session with 3-second wait for page load.

5.3 Flipkart Scraping

Target URL: Flipkart HP Pavilion search results page

- Navigates to Flipkart's HP Pavilion search results with a 4-second load wait.
- Uses Flipkart-specific CSS class names: jlJQ8S (product card), RG5Slk (name), hZ3P6wDeU9vF (price), MKiFS6 (rating).
- Iterates through product cards, stopping at 50 valid results.
- Uses try-except blocks throughout to gracefully skip unavailable fields.

6. Data Analysis Modules

6.1 Descriptive Statistics

Computed for both Price and Rating columns across the entire dataset and per platform. Metrics include count, mean, standard deviation, min, 25th percentile, median, 75th percentile, and max.

6.2 Confidence Interval Analysis

A 95% confidence interval is calculated for the mean price of each platform using SciPy's t-distribution (`stats.t.interval`). This provides a statistically grounded range within which the true mean price likely falls.

- Amazon: Mean price $\pm$ margin of error at 95% confidence.
- Flipkart: Mean price $\pm$ margin of error at 95% confidence.
- If confidence intervals overlap, platforms are considered to have similar pricing; otherwise, one platform is deemed consistently cheaper.

6.3 Standard Deviation Comparison

Price variability is measured for each platform. The platform with a lower standard deviation is identified as having more consistent and predictable pricing — a valuable signal for both consumers and sellers.

6.4 Independent Two-Sample T-Test

A two-sample independent t-test (`scipy.stats.ttest_ind`) is conducted on the price distributions of both platforms to determine whether any observed price difference is statistically significant ($p < 0.05$) or could be due to random sampling variation.

6.5 Price–Rating Correlation

The Pearson correlation coefficient between Price_Clean and Rating is calculated. Interpretation thresholds: $|r| < 0.3$ = very weak, $0.3\text{--}0.5$ = weak, > 0.5 = moderate to strong.

6.6 Price Segmentation

Each product is categorized into one of four market segments based on price:

- Budget: Under ₹50,000
- Mid-range: ₹50,000 – ₹65,000
- Premium: ₹65,000 – ₹80,000
- High-end: Above ₹80,000

Average customer rating per segment is also reported to surface value-for-money tiers.

6.7 Top Products Ranking

Three leaderboards are produced:

- Top 5 Most Expensive Products — ranked by Price_Clean descending.
- Top 5 Best Rated Products — ranked by Rating descending.
- Top 5 Best Value Products — ranked by Value_Score = (Rating / Price) × 10,000.

7. Data Visualizations

The project generates a comprehensive 2×3 subplot figure (hp_pavilion_analysis.png) using Matplotlib and Seaborn. Each panel targets a specific analytical question:

Panel	Chart Title	Purpose
[1,1]	Price Distribution by Platform	Histogram comparing price spread across Amazon and Flipkart with platform-specific colors (orange vs. blue).
[1,2]	Rating Distribution by Platform	Histogram showing how customer ratings are distributed on each platform, revealing rating concentration patterns.
[1,3]	Price Distribution — Box Plot	Side-by-side box plots exposing median prices, interquartile ranges, and outliers for each platform.
[2,1]	Price vs. Rating Scatter Plot	Scatter chart with trend line to visualize the direction and strength of the price-rating correlation.
[2,2]	Average Price by Platform	Bar chart with annotated values showing the mean listing price for Amazon vs. Flipkart.
[2,3]	Average Rating by Platform	Bar chart comparing mean customer satisfaction scores across both platforms.

8. Key Findings & Business Insights

8.1 Pricing Insights

Finding 1 — Noticeable Price Differences Across Platforms

The analysis identified visible pricing variations for similar HP Pavilion models between Amazon and Flipkart.

Root causes include platform-specific discount strategies, seller competition, inventory levels, and seasonal promotions.

Recommendation: Monitor platform-wise pricing in real time and align promotional calendars with platform audience behavior.

Finding 2 — Highly Competitive Pricing Environment

Both platforms maintain closely competing prices, indicating a mature and competitive e-commerce market for HP laptops.

Even small pricing changes can significantly influence consumer purchase decisions.

Recommendation: Implement continuous market monitoring and dynamic pricing mechanisms.

Finding 3 — Platform-Specific Pricing Strategies

One platform may consistently offer lower prices in certain product tiers, suggesting deliberate targeting of different customer segments.

Recommendation: Optimize product listings, pricing tiers, and promotional intensity separately per platform.

8.2 Rating & Satisfaction Insights

Finding 4 — Weak Positive Correlation Between Price and Rating

Statistical analysis found a weak positive correlation between price and customer rating.

Higher-priced laptops tend to receive marginally better ratings, but the relationship is not strong enough to be the primary driver.

Customer satisfaction is heavily influenced by performance, battery life, build quality, after-sales service, and reliability.

Recommendation: Focus on product quality and customer experience rather than relying solely on premium pricing.

Finding 5 — Mid-Range Segment Dominates Market

Mid-range HP Pavilion models (₹50k–₹65k) make up the largest segment with strong customer satisfaction scores.

Customers demonstrate clear preference for value-for-money over premium pricing.

Recommendation: Invest in feature optimization and marketing for mid-range products; target value-conscious buyers aggressively.

9. Project Output Files

File Name	Type	Description
Price_Comparison_Tool.ipynb	Jupyter Notebook	Main project code with all scraping, analysis, and visualization logic
hp_pavilion_price_comparison.csv	CSV Data File	Cleaned dataset of HP Pavilion listings from both platforms
hp_pavilion_analysis.png	PNG Image	2×3 subplot visualization dashboard saved at 150 DPI

10. Technical Skills Demonstrated

Programming & Tools - Python programming (Jupyter Notebook) - Selenium WebDriver automation - XPath selector engineering - ChromeDriverManager integration - Anti-bot-detection techniques	Statistical Analysis - Descriptive statistics (mean, std, min, max) 95% Confidence Interval estimation - Independent two-sample t-test - Pearson correlation coefficient - Value scoring and ranking
Data Engineering - Data cleaning and preprocessing (Pandas) - Handling missing and malformed values - Currency/string-to-numeric conversion - Dataset merging and filtering	Visualization & Communication - Multi-panel Matplotlib figure design - Seaborn styled charts - Annotated bar charts - Scatter plots with trendlines - Business insight documentation

- CSV export for reproducibility

11. Conclusion

This project successfully demonstrates how web scraping, data cleaning, exploratory data analysis, statistical testing, and visualization can be combined into a coherent, end-to-end market intelligence pipeline. By collecting live HP Pavilion laptop data from Amazon India and Flipkart, the tool provides actionable insights into platform pricing strategies, customer satisfaction trends, and product value positioning.

The analysis reveals several important market realities: the laptop e-commerce market is highly competitive; pricing differences between platforms are real but often modest; and customer satisfaction is driven by far more than price alone. Mid-range products dominate in both volume and satisfaction, indicating a strong market opportunity for value-focused positioning.

From a technical perspective, the project showcases industry-level capabilities including live web automation, statistical inference, and data-driven storytelling — skills directly applicable to roles in Data Analytics, Business Intelligence, and Market Research.

Future Scope
Real-time price tracking dashboard with automated daily scraping and alerts.
Extension to additional platforms (Croma, Vijay Sales, Reliance Digital) and product categories.
Machine learning models for price prediction and demand forecasting.
Interactive web app (Streamlit/Dash) for non-technical stakeholder access.
Sentiment analysis on customer reviews for deeper satisfaction modeling.

End of Document

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